Computer Vision

DPS920

Session 15

Overview

LR
Multiple LR
MSE, MAE
Examples

Agenda

Optimization
Gradient Descend
Example

What is Left?

1. Optimization and Loss Function
2. Logistic Regression
3. ML and Images
4. Perceptron and Neural Networks
5. Deep Neural Networks
6. Convolution Neural Networks (CNN)
7. Advanced CNNs
8. Introduction to object detection, segmentation, and image generation methods with AI
9. (Vision today, tools, future of CV)
10. Project

Problem

- Some datasets have millions of features -> impossible to calculate
- They are not always linear
- Some problems do not have closed form formulas

Problem

- Some datasets have millions of features -> impossible to calculate
- They are not always linear
- Some problems do not have closed form formals

Solution -> Optimization

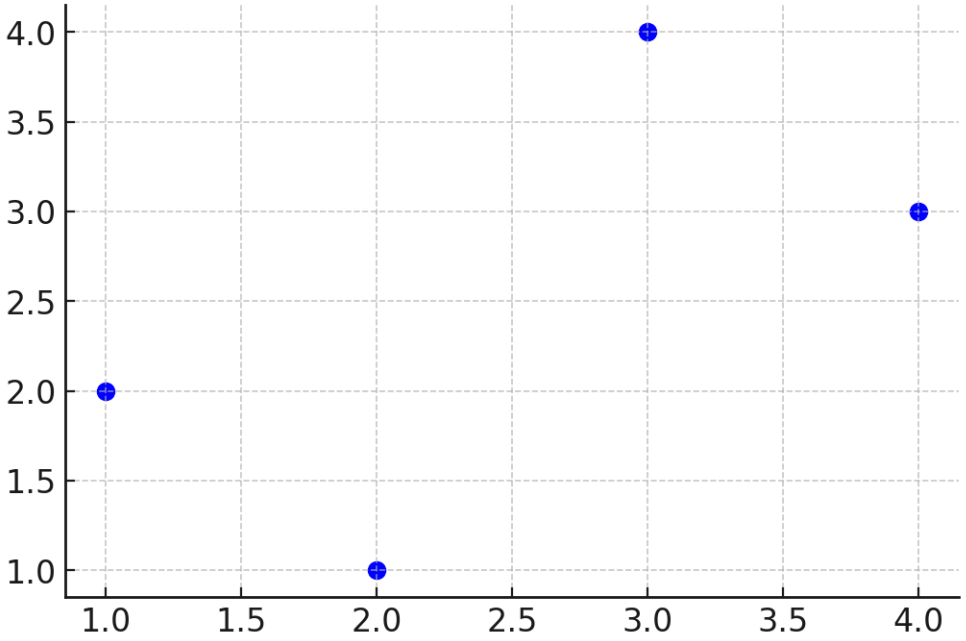
Optimization

Start random in space

Take gradual steps towards your goal

Not the best best answer but a solution close to the best

Example



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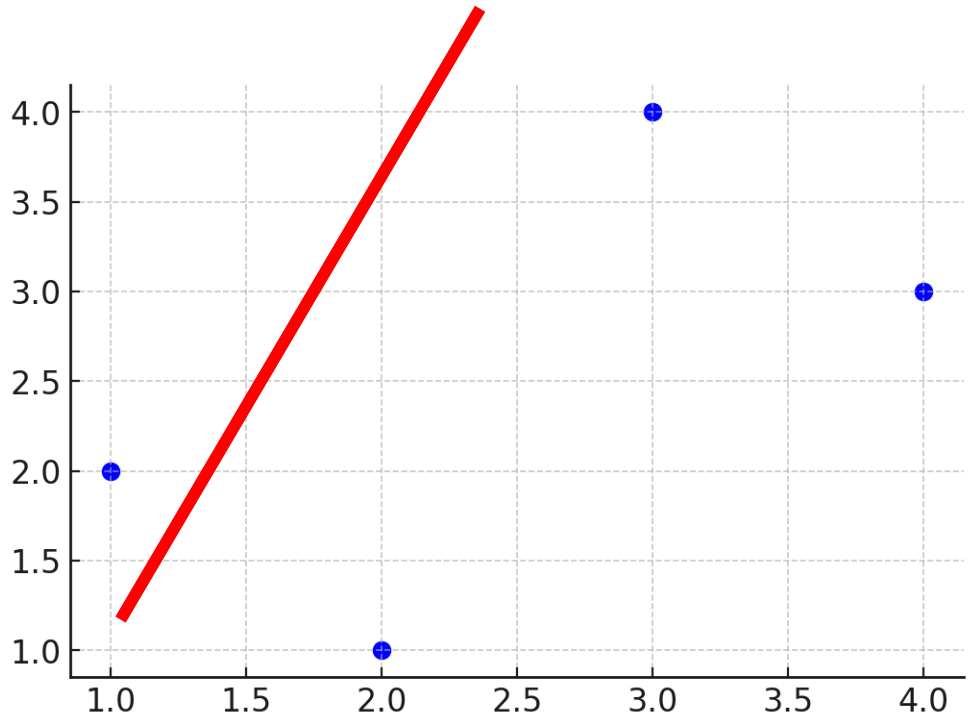


What is our goal?

What is our goal?

Minimize loss

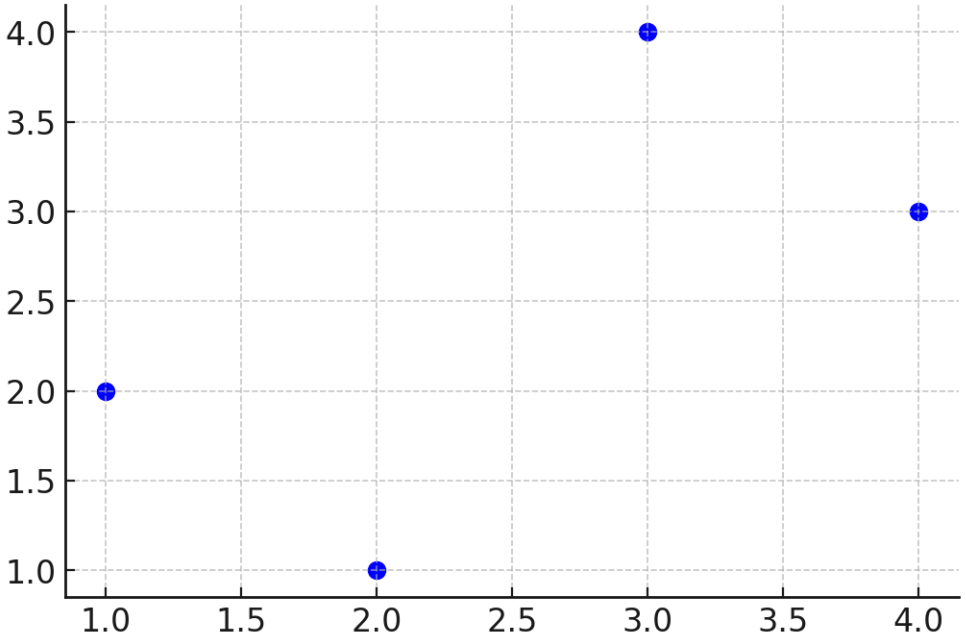
Gain parameter values

What Was Loss?

- MSE (mean squared error)

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Example



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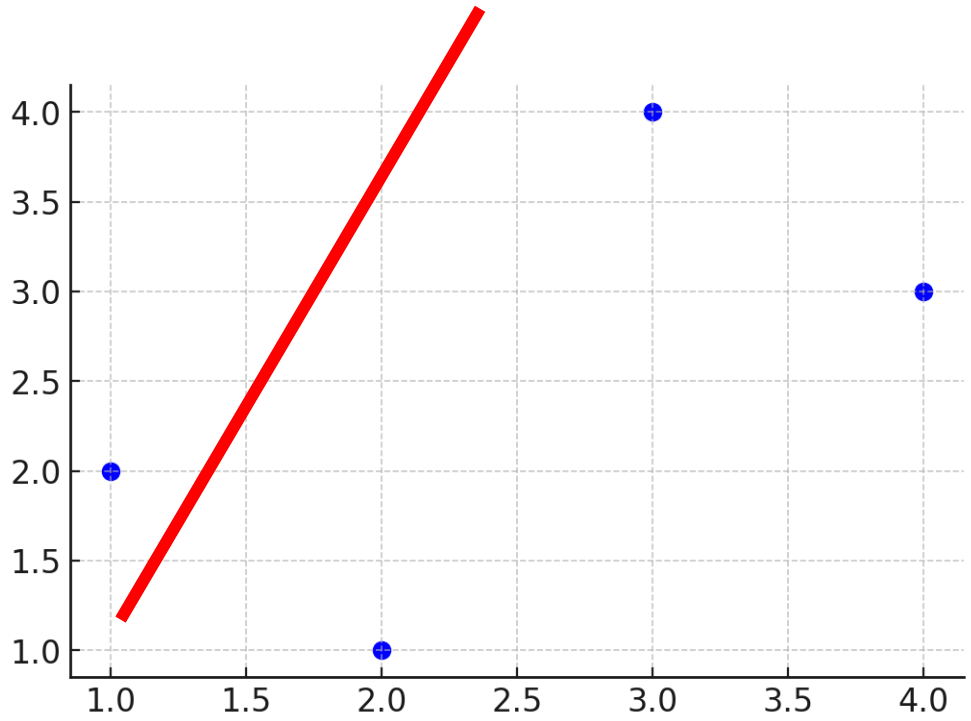

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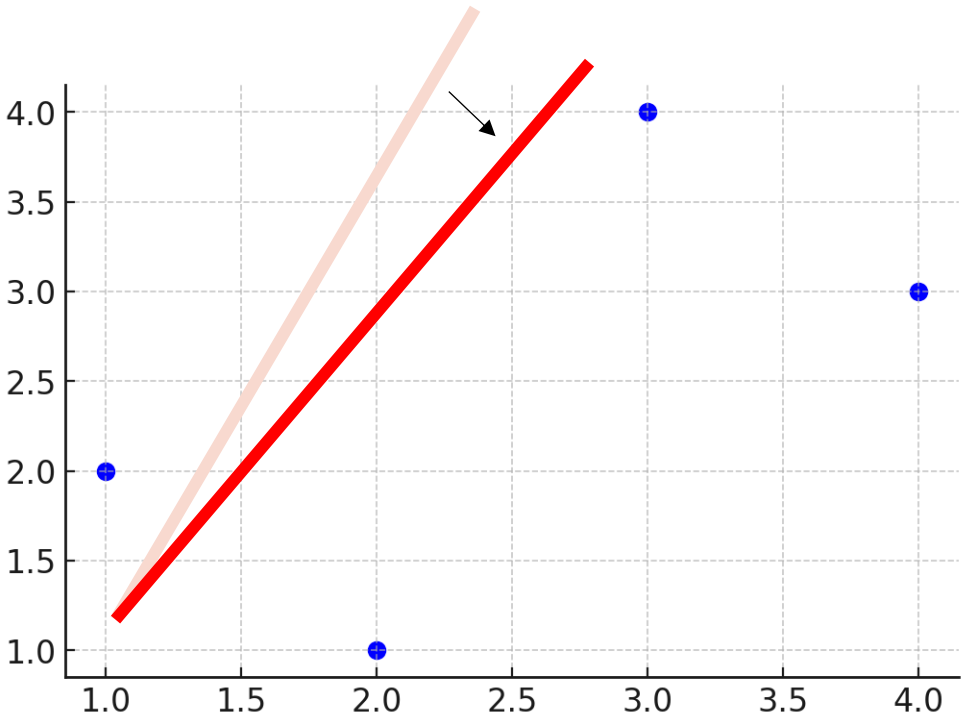

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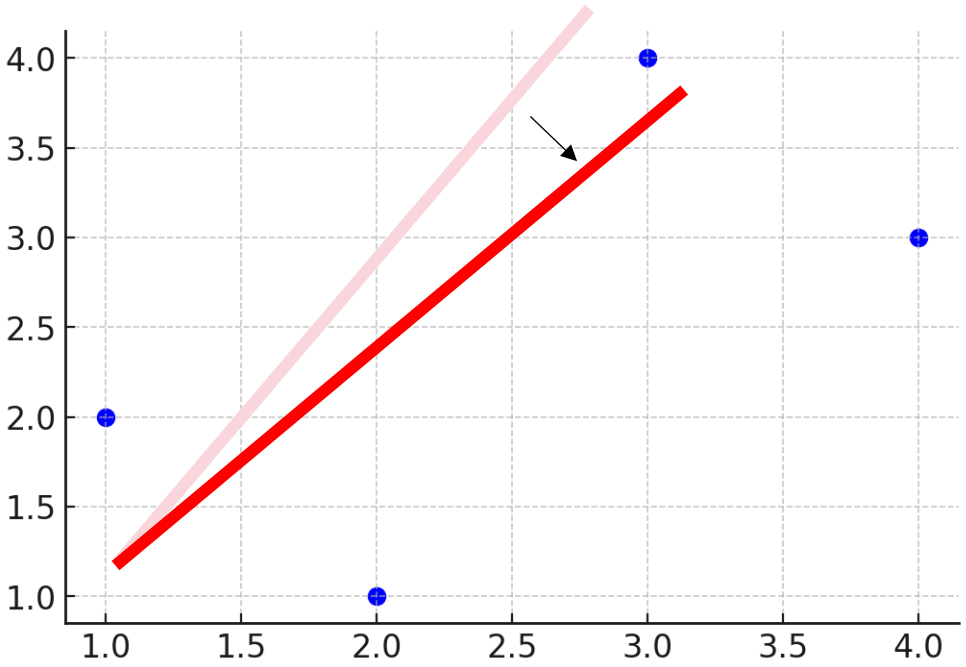

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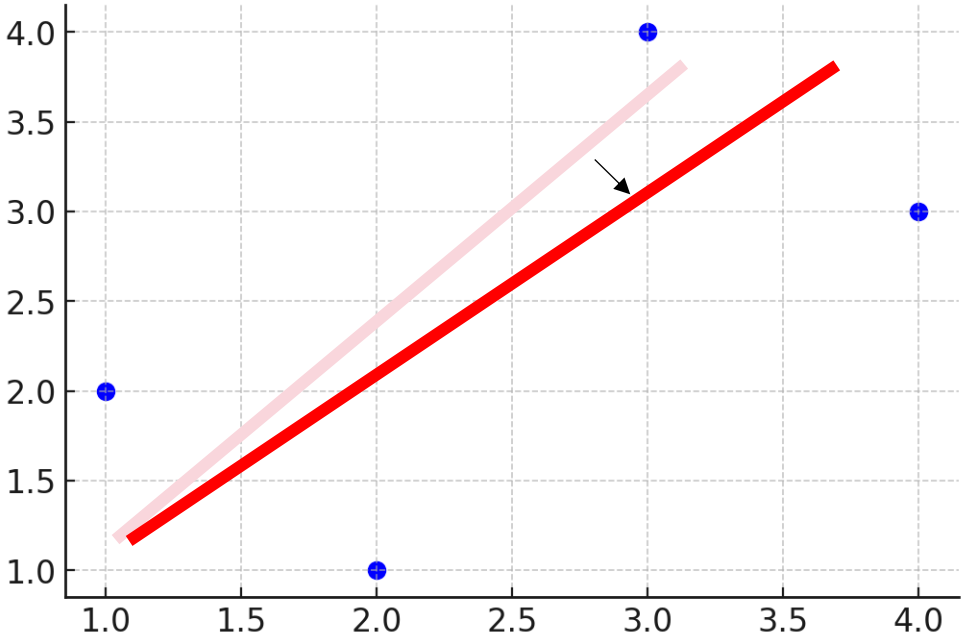
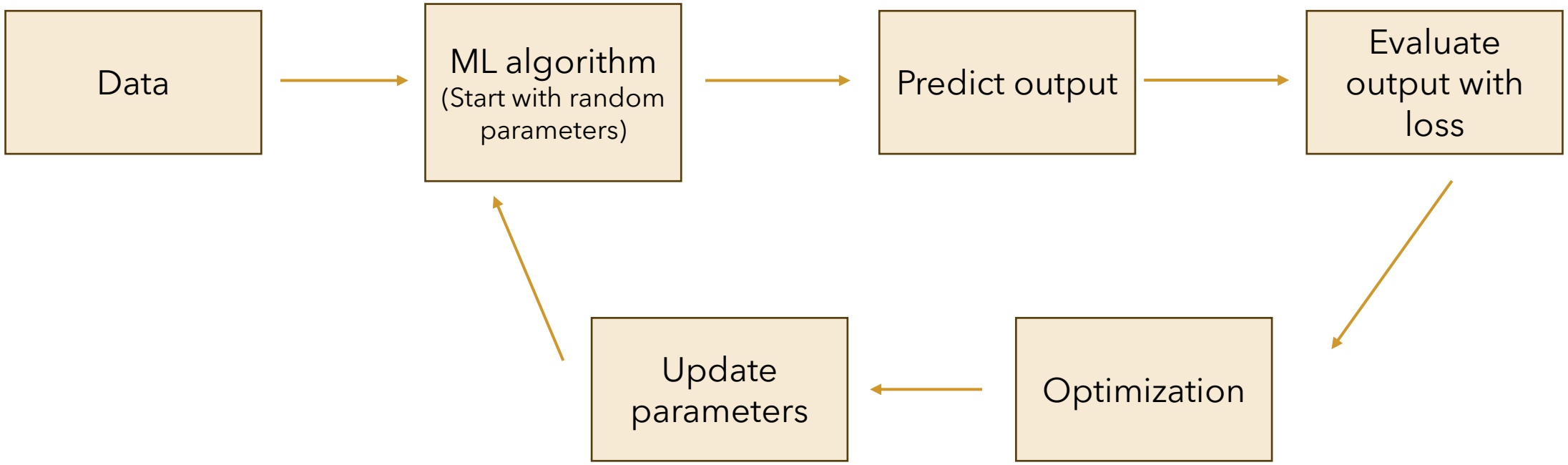


Framework


Data
ML algorithm
(Start with random parameters)
Predict output
Evaluate output with loss
Optimization
Update parameters

Optimization Algorithms

- Gradient Descend
- Stochastic Gradient Descend
- Adam
- AdamW
- RMSProp
- Newton’s Method

Gradient Descend

- The problem was how we get to update the line in LR

Mathematically How?

- We should solve this mathematically
- But how?

Gradient Descend

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- Gradient descend does this with calculating derivatives again!

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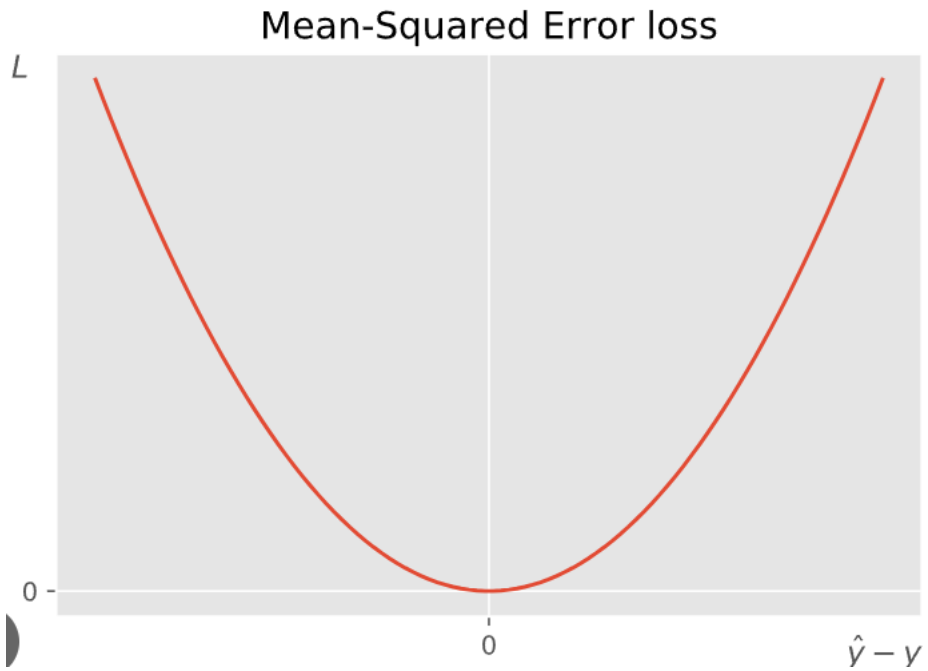
- It updates parameters by moving in the opposite direction of their derivatives with respect to loss!

But what do we mean?

opposite direction of parameter derivatives with respect to loss!

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$



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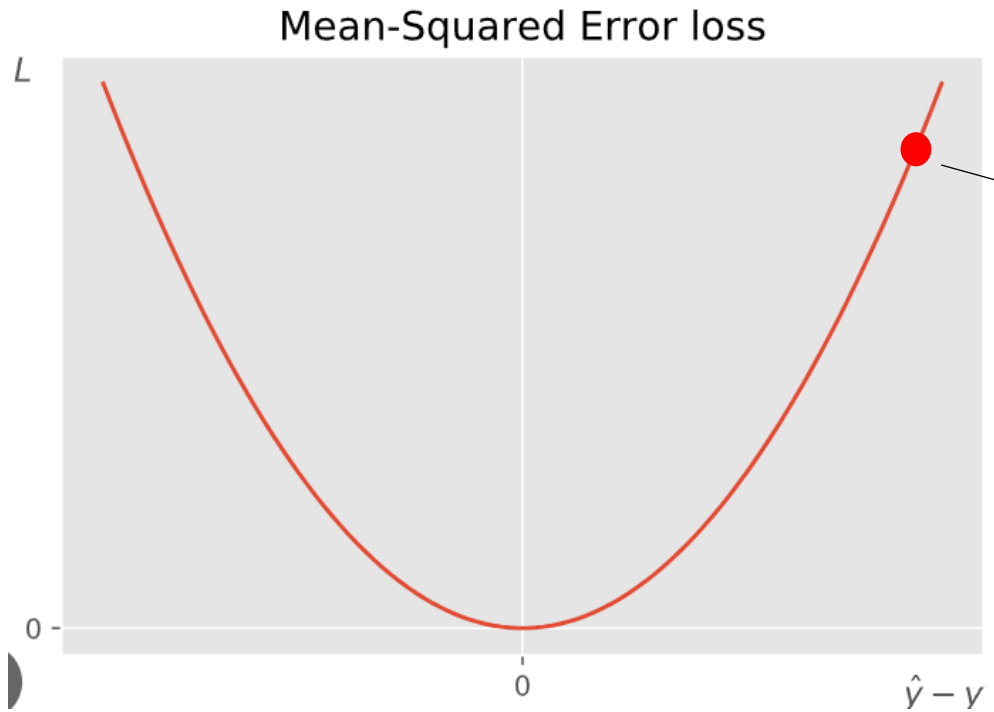
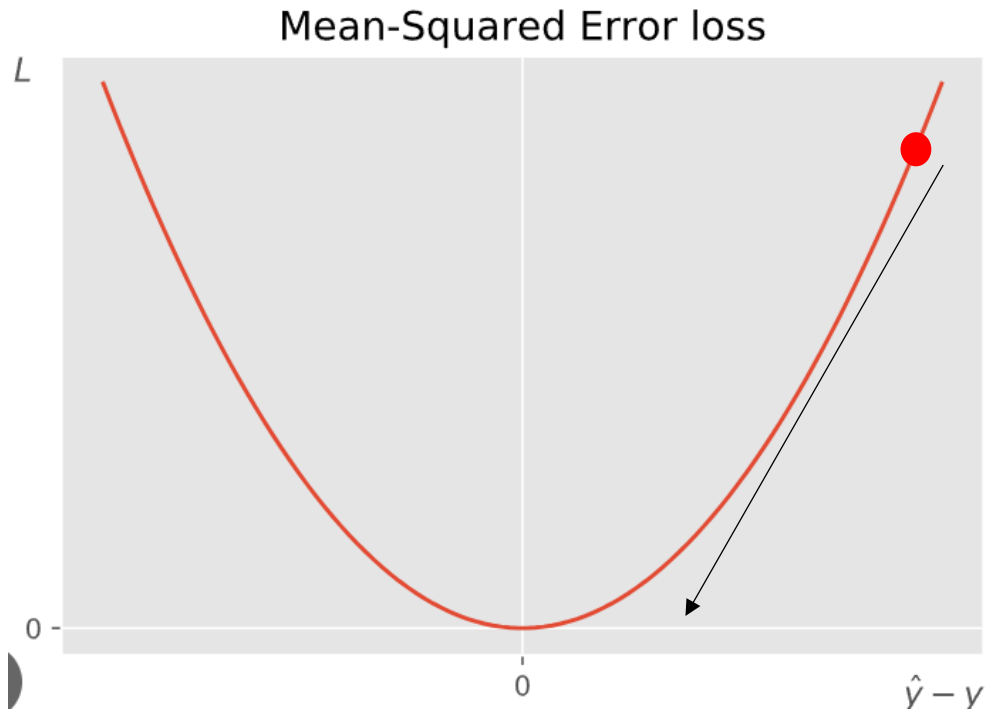


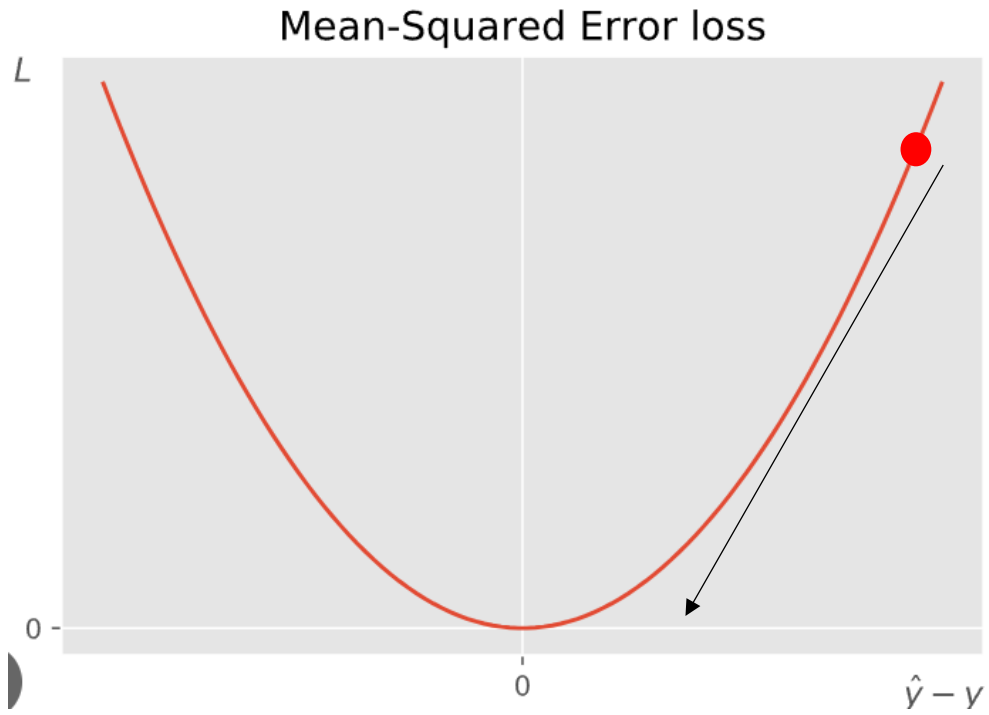
Our first average loss with the random line

opposite direction of parameter derivatives with respect to loss!



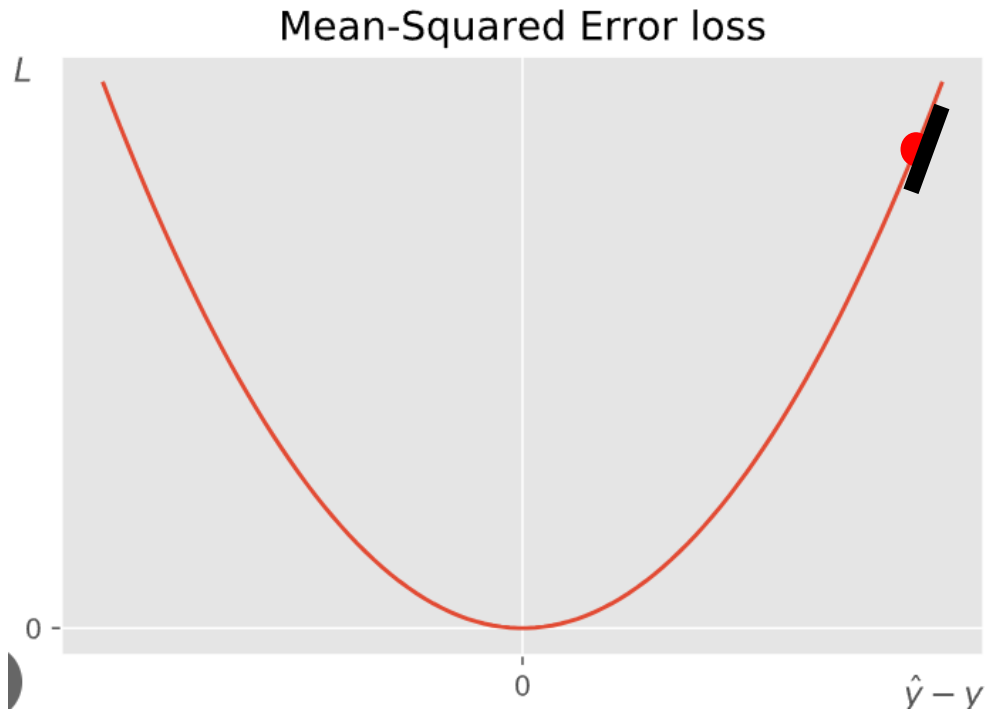
We want this loss to be as close as to 0

opposite direction of parameter derivatives with respect to loss!



In order to move towards that direction, we have to move in the opposite direction of slope (derivate)

opposite direction of parameter derivatives with respect to loss!



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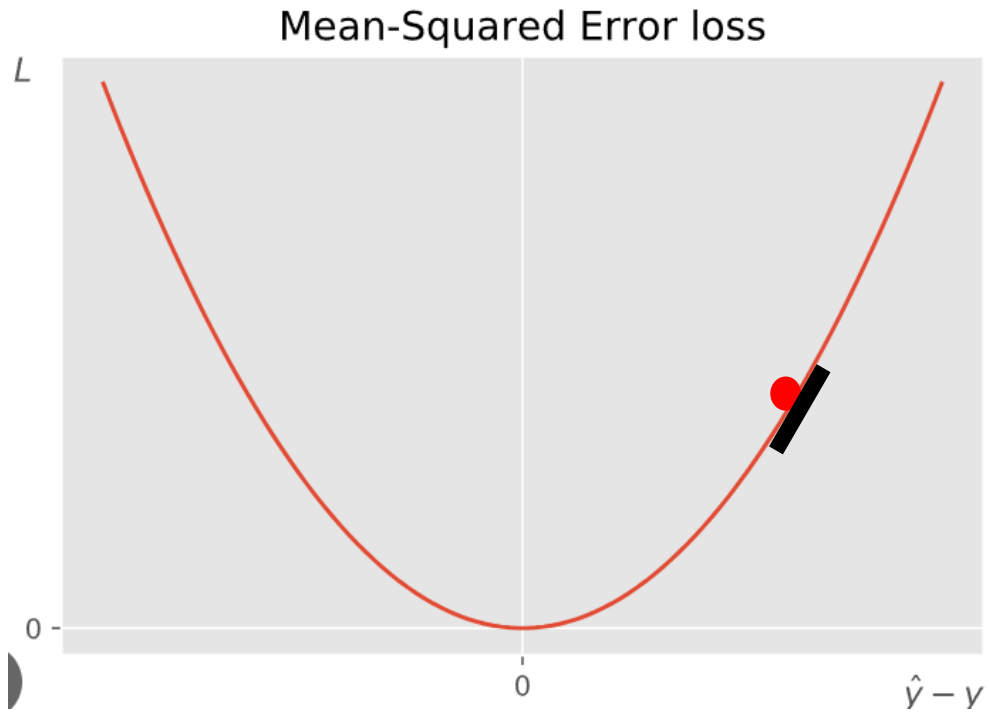

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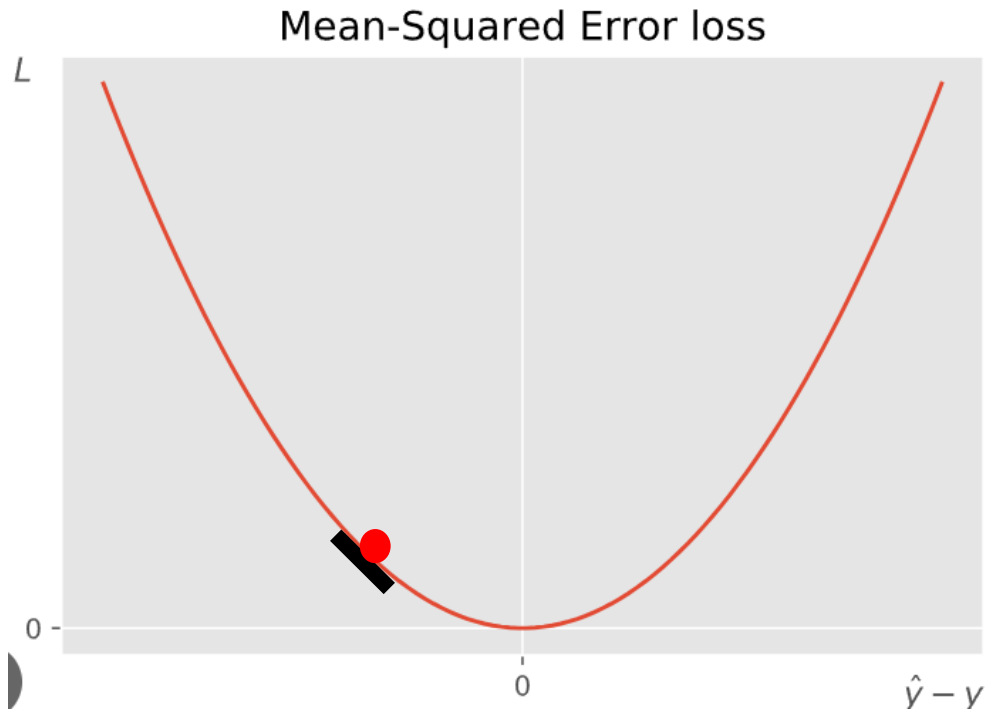
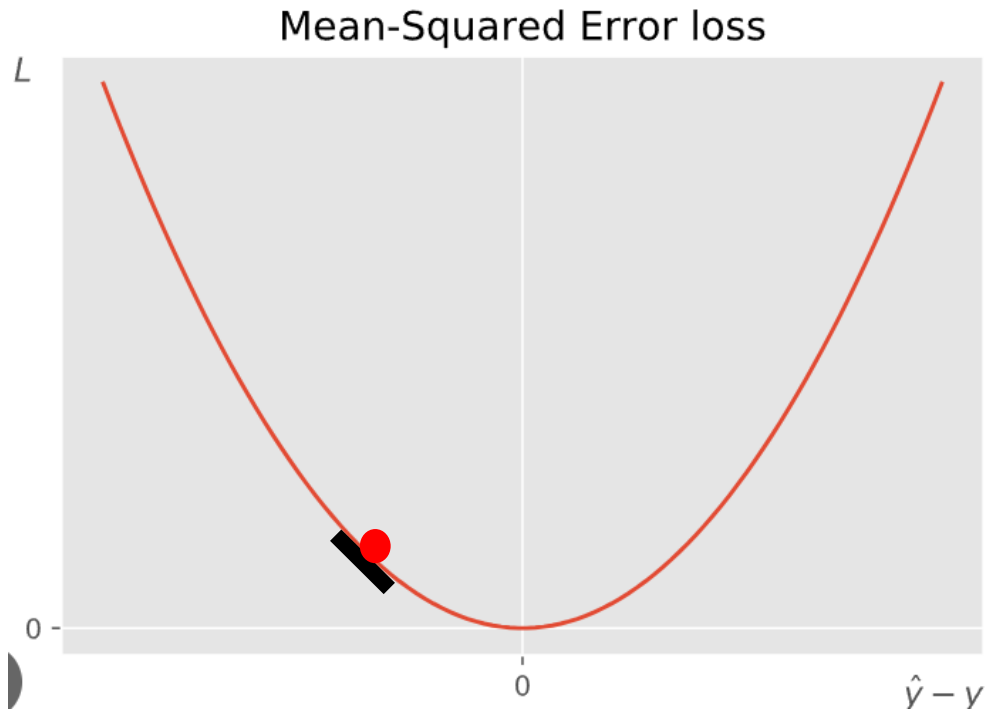


$$w^{+} = w^{-} - \frac{\partial L}{\partial w}$$

opposite direction of parameter derivatives with respect to loss!



$$w^{+} = w^{-} - \alpha \frac{\partial L}{\partial w}$$

Now let's see mathematically!

- opposite direction of **parameter derivatives** with respect to loss!
- Loss is calculated:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Now let's see mathematically!

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- Loss is calculated:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Given in data

ax+b

Now let's see mathematically!

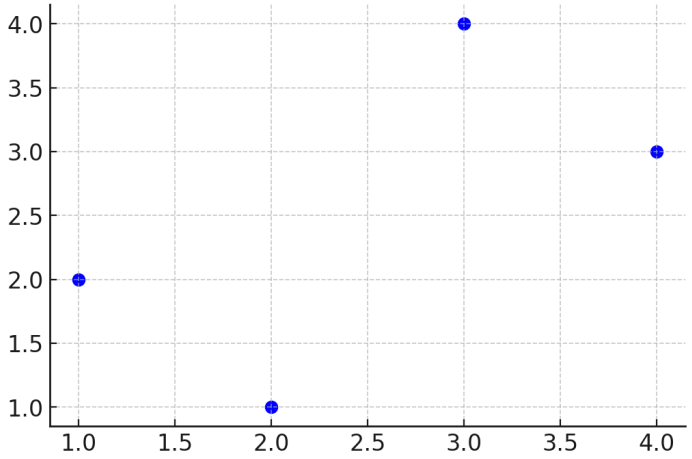
$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Given in data

2x+4

Now let's see mathematically!

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$



Data = (1,2), (2,1), (3,4), (4,3)
y′ = 2x + 4

Loss = [$(2-6)^2$+ $(1-8)^2$+ $(4-10)^2$+ $(3-12)^2$] / 4 = [16 + 49 + 36 + 81] / 4 = 45.5

Now let's see mathematically!

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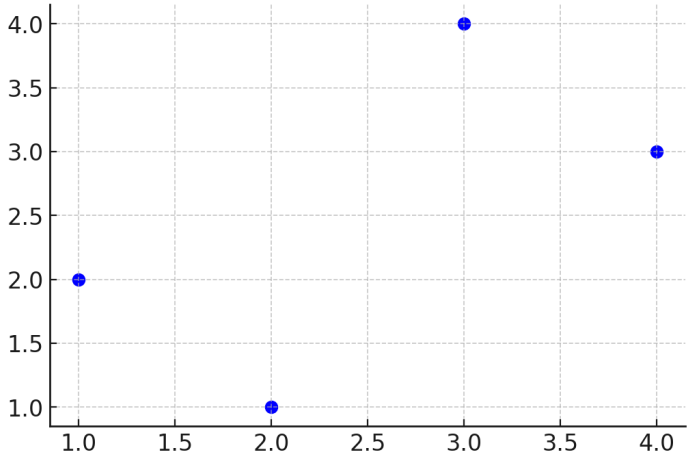

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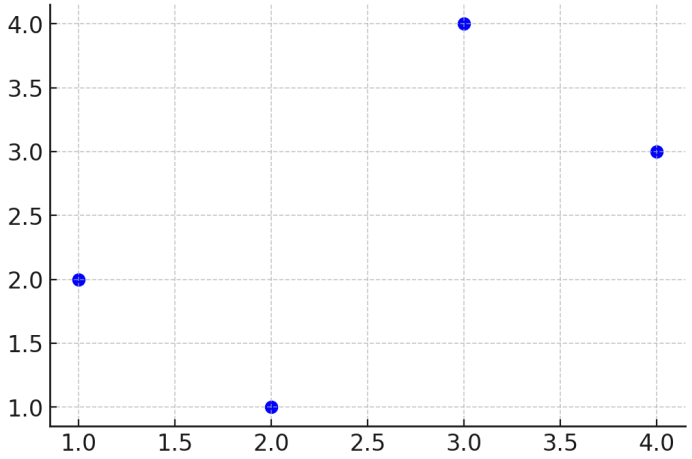
$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y - (mx + c))^2$$

Now let's see mathematically!

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$



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$$E = \frac{1}{n}\sum_{i=0}^{n}(y_i - (mx_i + c))^2$$

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$$D_m = \frac{1}{n}\sum_{i=0}^{n} 2(y_i - (mx_i + c))(-x_i)$$

$$D_m = \frac{-2}{n}\sum_{i=0}^{n} x_i(y_i - \bar{y}_i)$$

$$m = m - \alpha \times D_m$$

$$D_c = \frac{-2}{n}\sum_{i=0}^{n}(y_i - \bar{y}_i)$$

$$c = c - \alpha \times D_c$$

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$$D_m = \frac{-2}{n}\sum_{i=0}^{n} x_i(y_i - \bar{y}_i)$$

$$m = m - \alpha \times D_m$$

new m = 2 - 0.01 * 2 = 1.98

$$D_c = \frac{-2}{n}\sum_{i=0}^{n}(y_i - \bar{y}_i)$$

$$c = c - \alpha \times D_c$$

new c = 4 - 0.01 * 2 = 3.98

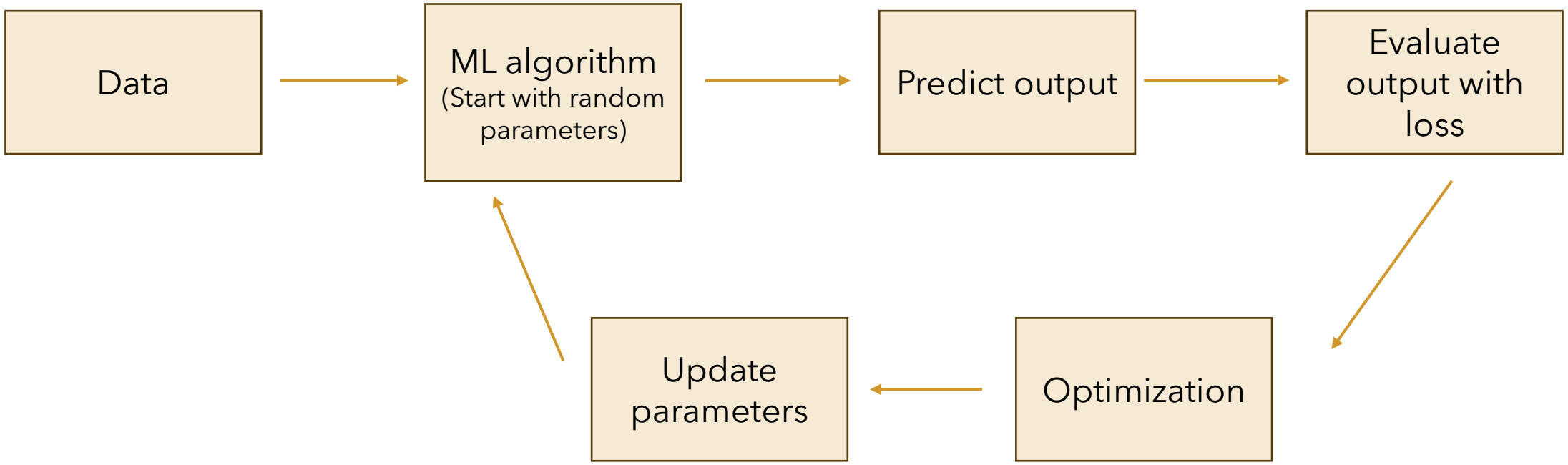
y = 1.98*x + 3.98

New line: $y = 1.98*x + 3.98$

Previous line: $y = 2x+4$

Best fitted line: $y = x + 0.6$

Framework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